

Client: Sydney Water Corporation
Project: Willoughby Submain Amplification Project
Location: Lawson Lane, Naremburn, NSW, 2065

Project No: 60764076
Logged by: VT
Start Date: 06/07/26

Checked by: RH
End Date: 15/07/26
Location Meth.: RTK

Driller: Terratest
Drill Rig: Comacchio GEO 405
Total Depth: 45.25 m

Hole Diameter: 96-300 mm
Inclination: 90°
Bearing: N/A

Easting: 333446.10
Northing: 6256299.70
Hor. Datum: MGA2020-56

RL: 84.4 m
Ver. Datum: AHD
Surface: Pavement (Asphalt)

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
D T				0.0			ASPHALTIC CONCRETE: dark grey, 20 mm thick, aggregate up to 10 mm.		FILL
		ES: 0.11-0.22 m PID=1.2 PPM D: 0.22-0.40 m ES: 0.35-0.40 m PID=1 PPM		80.0		CH	ASPHALTIC CONCRETE: dark grey, 90 mm thick, aggregate up to 30 mm. FILL - Sandy GRAVEL: fine to coarse grained, angular to sub-angular, dark brown and grey; sand, fine to coarse; with clay. Silty CLAY: high plasticity, brown.	D	RESIDUAL SOIL
VE				1.0		CI	Gravelly CLAY: medium plasticity, light brown; gravel, fine to medium, angular, shale.	w ≈ PL	
		SPT: 1.50-1.95 m 14.24.30 N=54 D: 1.50-1.95 m PP=400, 460, 390 kPa PP=>500, >500, >500 kPa		2.0		CL	Sandy CLAY: low plasticity, pale grey mottled brown; sand, fine to coarse grained; with gravel, fine to medium, sub-angular, shale; trace iron staining.	H	
		SPT: 3.00-3.21 m 30, 16/60 mm HB N=R D: 3.00-3.21 m PP=>500, >500, >500 kPa		3.0				w < PL	
U				4.0					
		SPT: 4.50-4.95 m 18, 35, 15 N=50 D: 4.60-4.95 m PP=500, 430, 460 kPa PP=>500, >500, >500 kPa		5.0		CL	Silty CLAY: low plasticity, dark brown, pale grey; with sand, fine to coarse grained; with gravel, fine to medium, angular, shale; iron staining.	w < PL	EXTREMELY WEATHERED MATERIAL
		SPT: 6.00-6.05 m 16/50 mm HB N=R		6.0				H	
				7.0		CH	Silty CLAY: medium to high plasticity, grey.	w ≈ PL	
		SPT: 7.60-7.65 m 12/50 mm HB N=R		8.0			Log continued on next page.		

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[illegible]

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METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x Is(50) A: • D: ○ UCS: □	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
HQ3	16.47	6	Is(50) D=0.6 MPa Is(50) A=1.27 MPa C: 16.70-17.00 m C: 17.03-17.23 m UCS=16 MPa	64.0		LAMINITE: siltstone (70%), dark grey, interlaminated at 0-5° with sandstone (30%), fine grained, pale grey.	SW	100 (98) [92]			15.94-16.00 m: EW, RF, PR, CL, 60 mm thick 16.18 m: P, 0°, RF, PR, CN	
						NO CORE: 16.22-16.47 m.					16.47-16.49 m: DB 16.57-16.85 m: DB, x 3	
						LAMINITE: siltstone (70%), dark grey, interlaminated at 0-5° with sandstone (30%), fine grained, pale grey.	SW				17.00 m: HB	
	18.15	7	Is(50) D=1.21 MPa Is(50) A=1.5 MPa	63.0		SANDSTONE: medium to coarse grained, pale grey, very thinly bedded at 20° with grey laminations .	MW / SW	97 (97) [82]			17.24 m: DB 17.29-17.32 m: SS, 0-5°, SM, UN, CI, 30 mm thick HAWKESBURY SANDSTONE 17.35-17.43 m: DB 17.52 m: J, 50°, RF, PR, CN, 110 mm persistence 17.60-17.84 m: DB, x 2	
											17.99-18.00 m: HB 18.13-18.25 m: DB, x 2	
											18.32 m: P, 5°, RF, PR, CN	
	21.15	8	Is(50) D=0.74 MPa Is(50) A=0.92 MPa	62.0				100 (99) [77]			18.75 m: J, 70°, RF, PR, CN, 270 mm persistence 18.88 m: P, 5°, RF, PR, CN 19.00 m: HB 19.17 m: J, 65°, RF, PR, CN, 100 mm persistence 19.35 m: P, 20°, RF, PR, VN CI	
						NO CORE: 19.23-19.32 m.					19.68 m: P, 15°, RF, PR, VN CI	
						SANDSTONE: medium to coarse grained, pale grey, very thinly to thinly bedded at 10-20° with grey laminations.	SW				20.00 m: HB	
	21.15	8	Is(50) D=1.49 MPa Is(50) A=1.74 MPa	61.0							20.38-20.39 m: DB 20.40-20.41 m: EW, 5°, RF, PR, CI, 10 mm thick 20.67 m: P, 5°, RF, PR, VN CI 20.85 m: P, 5°, RF, PR, CN	
											21.00 m: HB 21.12 m: J, 75°, RF, PR, healed, 170 mm persistence 21.19 m: DB 21.23 m: J, 70°, RF, PR, CT CI, 130 mm persistence 21.30 m: P, 5°, RF, PR, CN 21.55-21.70 m: P, 10°, RF, PR, CN, x 2 21.64 m: J, 70°, RF, PR, CN, healed, 100 mm persistence	
						LAMINITE: sandstone (70%), pale grey, fine grained, interlaminated at 10-20° with siltstone (30%), dark grey.	MW				21.95 m: DB 22.00 m: HB	
	21.15	8	Is(50) D=0.85 MPa Is(50) A=1.67 MPa	59.0		SANDSTONE: medium to coarse grained, pale grey, very thinly to thinly bedded at 10-20° with grey laminations.	SW				22.30 m: DB	
											22.52-22.54 m: DB 22.58-22.69 m: P, 5-15°, RF, PR, CN, x 2 22.63 m: J, 10°, RF, PR, CN	
											23.00 m: HB	
	21.15	8	Is(50) D=1.22 MPa Is(50) A=1.05 MPa	58.0							23.37-23.57 m: P, 0-5°, RF, PR, CN, x 5	
											23.90-23.94 m: CS, RF, PR, SP, 40 mm thick	
21.15	8	Is(50) D=0.73 MPa Is(50) A=1.8 MPa	57.0									

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METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x Is(50) A: • Is(50) UCS: □	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
HQ3	24.15		Is(50) D=0.73 MPa Is(50) A=1.07 MPa	56.0			SANDSTONE: medium to coarse grained, pale grey, very thinly to thinly bedded at 10-20° with grey laminations.	SW	98 (98) [86]			24.00 m: J, 60°, RF, PR, CN, 110 mm persistence 24.10 m: J, 70°, RF, PR, CN, 70 mm persistence 24.15 m: J, 55°, RF, PR, CN 24.31 m: P, RF, PR, CN 24.35 m: P, RF, PR, CN 24.60 m: J, 65°, RF, PR, CN, 90 mm persistence 24.71 m: J, 30°, RF, PR, CN, discontinuous 24.84 m: DB 24.92 m: P, 10°, RF, IR, CT X 25.46 m: J, 45°, RF, PR, CN, 60 mm persistence 25.67 m: J, 65-75°, RF, PR, CN, 130 mm persistence 25.71 m: J, 60°, RF, PR, CN, 90 mm persistence 25.86 m: P, 20°, RF, PR, CN 25.99-26.00 m: HB 26.29 m: P, 20°, RF, PR, CN 26.51 m: P, 20°, RF, PR, CN 26.75 m: P, 20°, RF, PR, CN 27.00 m: HB
	27.15		Is(50) D=1.22 MPa Is(50) A=1.51 MPa C: 25.88-25.99 m C: 26.00-26.26 m UCS=25 MPa Is(50) D=1.49 MPa Is(50) A=1.63 MPa	55.0				SW to FR				
				25.0								
HQ3				54.0								
				53.0								
				52.0								
HQ3				51.0								
				50.0								
				49.0								
HQ3				48.0								
				47.0								
				46.0								
HQ3				45.0								
				44.0								
				43.0								
HQ3				42.0								
				41.0								
				40.0								
HQ3				39.0								
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HQ3				36.0								
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HQ3				6.0								
				5.0								
				4.0								
HQ3				3.0								
				2.0								
				1.0								
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HQ3	12	33.29	C: 32.02-32.22 m UCS=29 MPa C: 32.22-32.38 m Is(50) D=1.32 MPa Is(50) A=1.31 MPa	48.0			SANDSTONE: medium to coarse grained, pale grey, very thinly to thinly cross-bedded at 0-25° with grey laminations, trace siltstone clasts up to 20mm thick.	SW to FR				32.00 m: HB 32.03 m: P, 10°, RF, PR, VN CI
			Is(50) D=0.86 MPa Is(50) A=1.09 MPa	47.0					100 (100) [100]			32.92-32.96 m: P, 10°, RF, PR, CN, x 2 33.00 m: HB 33.30 m: DB
HQ3	13	36.33	Is(50) D=0.91 MPa Is(50) A=0.93 MPa	46.0								34.00 m: HB 34.34 m: P, 10°, RF, PR, CN 34.61 m: P, 0-5°, RF, IR, CN
			Is(50) D=1 MPa Is(50) A=0.74 MPa	45.0			from 34.80 m: becoming pale orange grey, trace fine quartz gravel inclusions, iron staining					35.00 m: HB 35.45 m: P, 0°, RF, PR, CT SP 36.00 m: HB
HQ3	13	39.38	Is(50) D=0.68 MPa Is(50) A=0.75 MPa	44.0					100 (100) [100]			36.33 m: P, 0-5°, RF, IR, CT X 36.46 m: DB
			Is(50) D=2.03 MPa Is(50) A=1.92 MPa C: 37.69-37.87 m UCS=31 MPa C: 37.87-37.98 m	43.0								36.99-37.00 m: HB 37.13 m: DB 37.28 m: J, 40°, RF, PR, CN 37.49 m: J, 45°, RF, PR, CN 37.69 m: P, 5°, RF, PR, SN Fe 37.87 m: P, 5°, RF, PR, CN 37.99 m: HB 38.00 m: HB
HQ3	13	39.38	Is(50) D=1.32 MPa Is(50) A=1.46 MPa	42.0								38.28 m: P, 5°, RF, PR, CT CI 38.38 m: P, 5°, RF, PR, CT CI
			Is(50) D=1.53 MPa Is(50) A=1.78 MPa	41.0					100 (99) [93]			38.72 m: DB 38.99-39.00 m: HB

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HQ3	14		C: 40.00-40.09 m C: 40.09-40.29 m UCS=27 MPa Is(50) D=1.08 MPa Is(50) A=1.26 MPa C: 41.00-41.35 m Is(50) D=1.31 MPa Is(50) A=1.13 MPa C: 41.40-41.66 m C: 41.70-42.00 m UCS=25 MPa Is(50) D=0.43 MPa Is(50) A=0.38 MPa	40.0	41.0	42.0	SANDSTONE: medium to coarse grained, pale grey, very thinly to thinly cross-bedded at 0-25° with grey laminations, trace siltstone clasts up to 20mm thick. from 41.90 m: becoming pale orange brown SANDSTONE: medium to coarse grained, pale grey, indistinctly very thinly bedded at 0-5°. SILTSTONE: dark grey, thinly laminated at 0-5°.	SW to FR HW XW MW to SW HW MW HW SW to FR				40.00 m: HB 40.08-40.42 m: DB, x 5 40.74 m: J, 75°, RF, PR, CN, 120 mm persistence 41.00 m: HB 41.40 m: P, 10°, VR, PR, X 41.71 m: P, 0°, RF, PR, SN Fe 42.00 m: HB 42.02 m: P, 10°, RF, PR, SN Fe 42.08 m: J, 30°, RF, PR, CT SC 42.28 m: DB 42.33-42.35 m: EW, 0-10°, CU, Cl, 20 mm thick 42.50 m: DB 42.67 m: J, 80°, RF, PR, healed, discontinuous, 210 mm persistence 42.77 m: P, 0°, SM, IR, CT Cl 42.87 m: P, 30°, SM, IR, CN 43.00 m: J, 15°, RF, PR, CN 43.22 m: DB 43.37 m: J, 50°, SM, PR, CT Cl, 60 mm persistence 43.80 m: DB 44.00 m: HB 44.03 m: DB 44.27-45.08 m: DB, x 3 45.00 m: HB 45.10 m: DB
	15		Is(50) D=0.93 MPa Is(50) A=1.53 MPa	35.0	46.0	47.0	Terminated at 45.25 m. Vibrating wire piezometer installed.					

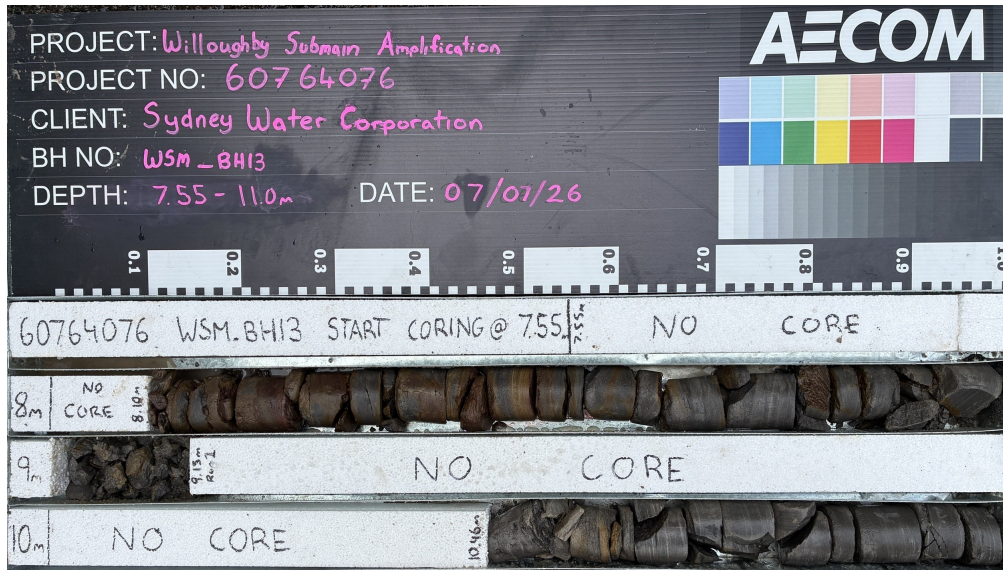
Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

CORE PHOTO REPORT

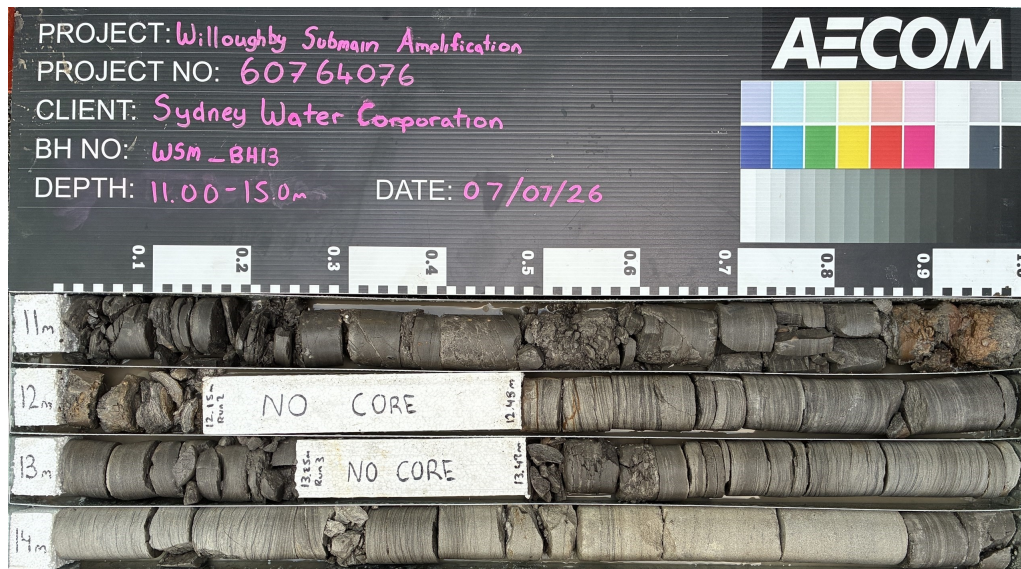
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Core Tray - 7.55 to 11.00 m



Core Tray - 11.00 to 15.00 m



Core Tray - 15.00 to 19.00 m

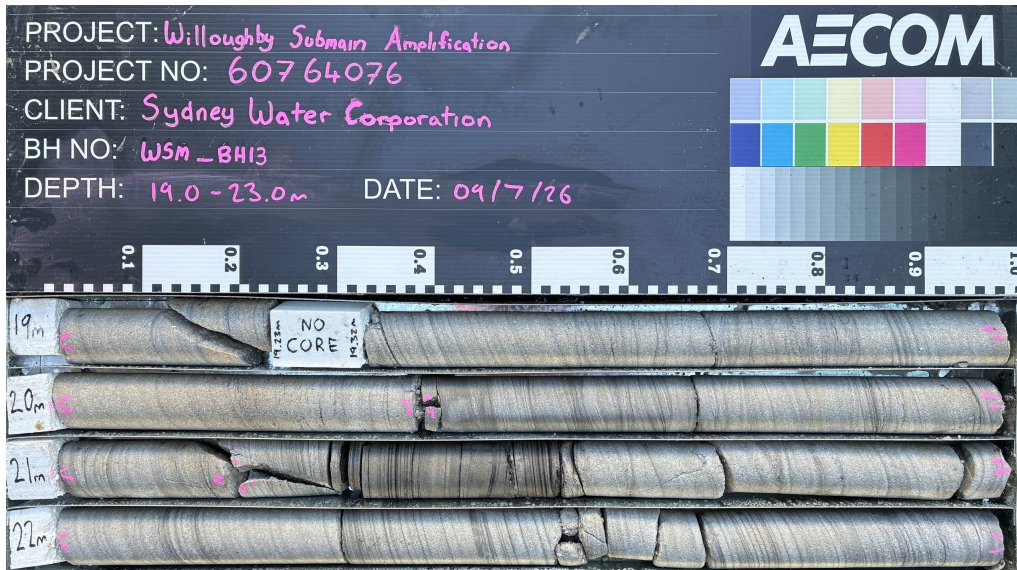
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Core Tray - 19.00 to 23.00 m



Core Tray - 23.00 to 27.00 m



Core Tray - 27.00 to 31.00 m

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CORE PHOTO REPORT

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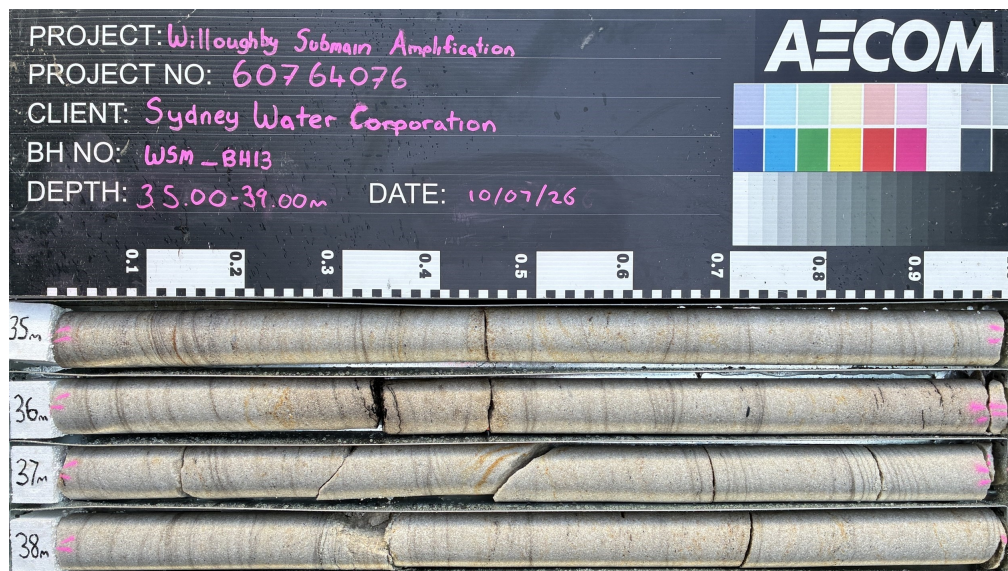
PROJECT NO. : 60764076

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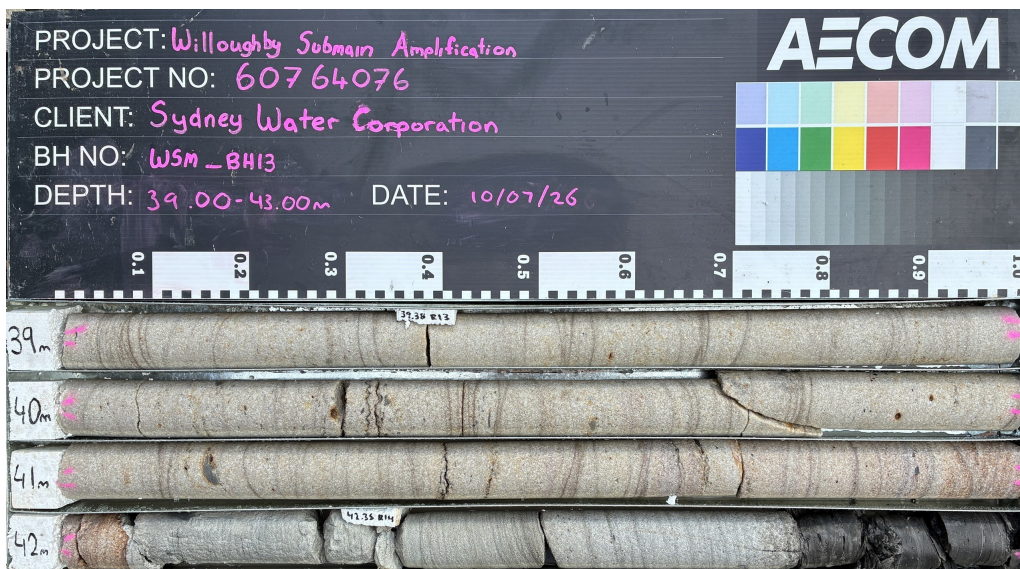
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Core Tray - 31.00 to 35.00 m



Core Tray - 35.00 to 39.00 m



Core Tray - 39.00 to 43.00 m

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CORE PHOTO REPORT

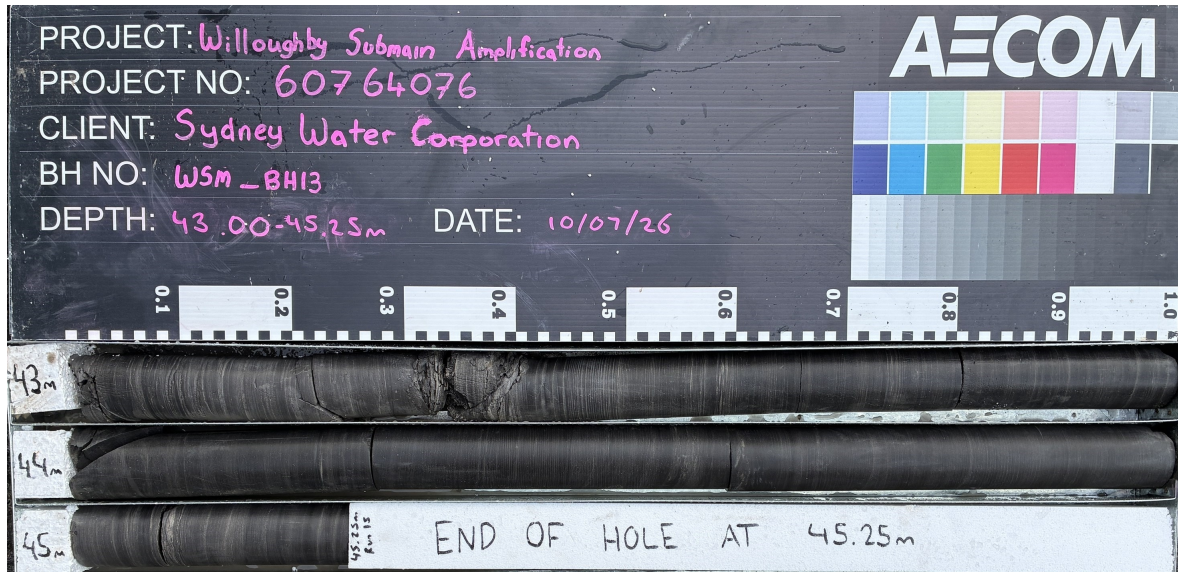
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